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Multiplicity Control in Clinical Trials with Adaptive Selection Followed by Group-Sequential Testing

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ABSTRACT
We propose a strategy for managing the issue of multiplicity in clinical trials with adaptive selection followed by group-sequential testing. The approach employs a two-stage design and addresses clinical trials with multiple hypotheses. The first stage adaptively selects a subset of hypotheses for further testing, and the second stage monitors the remaining hypotheses based on group-sequential procedures. We provide a rigorous framework for controlling the overall Type I error rate across both stages, using group-sequential *p*-values and the closed testing principle to ensure statistical validity in the adaptive setting. Through a simulation study based on an oncology trial example, we demonstrate the effectiveness of the proposed method in controlling Type I error rate while maintaining sufficient power.

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Adaptive design; Enrichment design; Group-sequential *p*-values; Treatment selection

1. Introduction

Clinical trials often have multiple objectives for different endpoints, populations, and treatment groups (Bauer 1991; Bender and Lange 2001; Chow and Chang 2008; FDA 2017). Without enough data at the start of a trial, trialists often face difficult decisions to narrow the list of objectives so that only a manageable number of treatment arms are left with a clearly defined target population.

Over the past two decades, adaptive clinical trial designs have proven valuable in drug development, contributing to numerous regulatory approvals of new medical products (Sydes et al. 2012; Chen, Gesser, and Luxembourg 2015; Filozof et al. 2017; Mehta, Liu, and Theuer 2019; Suchting et al. 2021; Jones et al. 2022). These designs enhance decision-making by allowing researchers greater flexibility to analyze data during the trial and implement strategic modifications. When interim results indicate that certain treatments show less favorable benefit-risk profiles or that specific patient populations respond poorly to treatments, researchers can remove these treatment groups or populations from the study. This targeted approach leads to more streamlined and efficient clinical development programs.

Another important but complex aspect of adaptive designs is trial monitoring after treatment and/or population selection (Mehta et al. 2009; Chow and Song 2017; Park, Thorlund, and Mills 2018; Simon and Simon 2013; Van and Gail 2019; Thall 2021). This is especially important when clinical trials require long-term follow-up, for example, in oncology, where interim analyses of cumulative data from the selected treatment group(s) or patient population(s) can provide valuable assessments.

Despite their advantages, adaptive designs also raise practical challenges. Interim decisions are often made when follow-up is incomplete and long-term time-to-event endpoints remain immature, particularly in trials with continuous enrollment. As a result, interim analyses may rely on short-term or surrogate endpoints, and operational issues, such as enrollment overrun from discontinued hypotheses, may arise. These considerations underscore important tradeoffs between statistical efficiency and operational feasibility and motivate the development of methods that remain valid under realistic trial conditions.

As we will illustrate in the following sections, our approach begins with multiple hypotheses and consists of two stages in the design. In the first stage, some hypotheses are either dropped or selected, a process we refer to as the adaptive stage. The remaining hypotheses are monitored in the second stage based on group-sequential procedures. We aim to make this approach as general as possible, allowing selection based on the primary endpoint, secondary endpoints, and/or surrogate endpoints, as long as it maintains the independent increment property, that is, selection cannot be made based on future data.

This paper aims to develop a valid statistical method for controlling the overall Type I error rates arising from both adaptive selection and post-selection group-sequential testing.

2. Motivation Example

The methodology development is motivated by the design of an oncology clinical trial where treatment and population selection occur at the end of Stage-1, and a plan for group-sequential testing will be carried out during Stage-2 post-selection of treatment and population. The trial evaluates two experimental treatments, T_1 and T_2 versus the standard of care T_0, in the biomarker-positive (BM+) population and the overall population, based on two primary endpoints: progression-free

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Figure 1. Selection criteria based on objective response rate (ORR) at IA1.

survival (PFS) and overall survival (OS). To increase feasibility, the timing of interim analyses is driven by the information accumulated for the comparisons between T_2 and T_0 in the BM+ population.

Treatment and population selection will be conducted at the first interim analysis (IA1) based on objective response rate (ORR), a short-term endpoint. The selection criteria are presented in Figure 1 with details as follows:

S-1. If the difference in ORR between T_2 and T_0 in the BM+ population is less than 15%, stop everything;
S-2. Else, if both the difference in ORR between T_2 and T_0 in the overall population is less than 10%, and the difference in ORR between T_1 and T_0 in the BM+ population is less than 10%, stop the recruitment of both BM- population and T_1 arm (i.e., enrich the BM+ population in T_2 and T_0);
S-3. Else, if the difference in ORR between T_2 and T_0 in the overall population is less than 10%, but the difference in ORR between T_1 and T_0 in the BM+ population is greater than 10%, stop the recruitment of the BM- population (i.e., enrich the BM+ population);
S-4. Else, if the difference in ORR between T_1 and T_0 in the BM+ population is less than 10%, but the difference in ORR between T_2 and T_0 in the overall population is greater than 10%, stop the T_1 arm (i.e., select treatment T_2).
S-5. Otherwise, no enrichment and keep all treatment and biomarker subgroups.

Here, we assume that the selection process does not involve any formal hypothesis testing and is based only on descriptive statistics of ORR. At the outset of the trial, there are 8 hypotheses based on the PFS or OS, considering two treatment groups, two populations, and two primary endpoints (see Table 2 in Section 4 for details). The selection process at IA1 will eliminate some of these hypotheses and we list all scenarios in Figure 1. For example, if recruitment from the BM- population is stopped, there will be four hypotheses left (T_2 and T_1 vs. T_0 with two endpoints in the BM+ population); if the T_1 arm is terminated, there will be four hypotheses left (T_2 vs. T_0 in two populations with two endpoints); if only the treatment T_2 and the BM+ population are selected, only two hypotheses will remain (T_2 vs T_0 with two endpoints in the BM+ population).

According to the given selection criteria, if the study continues after the selection stage, T_2 and T_0 in the BM+ population will always continue at a minimum. Note that the selection criteria can be adapted based on specific study objectives, the motivating example is provided for illustrative purposes. In the subsequent group-sequential stage, three additional analyses will then be conducted according to the event numbers of T_2 vs T_0 in the BM+ population:

IA2. Interim analysis for PFS when 75% of the target number of PFS events has been observed;
IA3. Final analysis for PFS and interim analysis for OS when 100% of the target number of PFS events has been achieved;
FA. Final analysis for OS when 100% of the target number of OS events has been accrued. PFS analyses will be updated but will not be considered in multiplicity control.

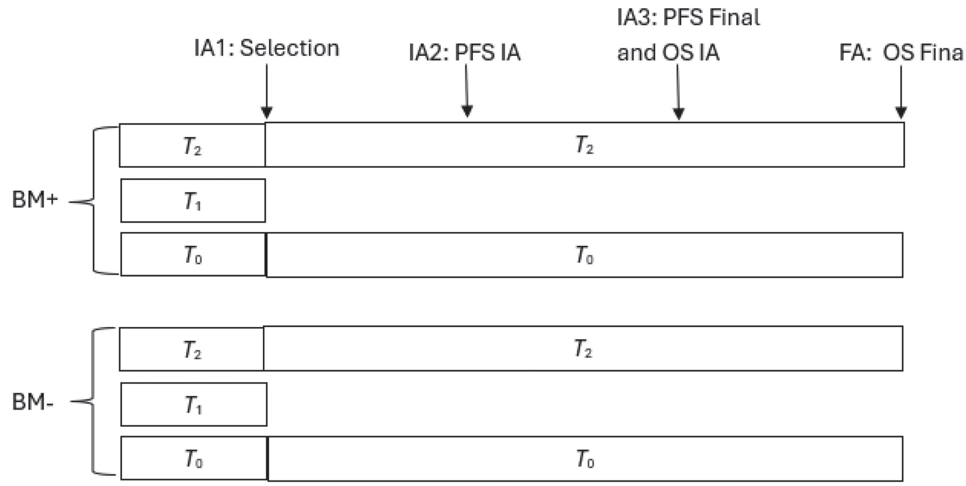

Figure 2. Schema if treatment T_1 is dropped.

Figure 2 presents a design schema for the scenario where the treatment arm T_1 is dropped after IA1. By convention, the dropped hypotheses will no longer be tested, and claims will be made only based on the testing results of the remaining hypotheses. To proceed, it is important to account for the trial design's adaptive and group-sequential features when testing the remaining hypotheses.

In the next section, we will propose a strategy to manage the multiplicity issues of this trial design, considering the selection process and the group-sequential testing of the (remaining) multiple hypotheses to achieve strong control of the family-wise Type I error rate. In addition, it is worth highlighting that the motivating example is provided for illustrative purposes, and its primary purpose is to demonstrate the proposed design framework. The proposed methods are flexible and can accommodate categorical, continuous, or time-to-event endpoints, as well as different hypothesis selection criteria and other operational characteristics, depending on the specific study objectives. This example represents a common setting in adaptive clinical trials for oncology drug development, where the primary objective is to evaluate treatment benefit in the biomarker-positive subgroup. In practice, trends observed in the biomarker-negative population could be incorporated as an additional futility criterion, depending on the trial objectives and clinical context.

3. Group-Sequential Tests after Adaptive Design

For simplicity, we assume that the adaptive stage (Stage-1) only includes one descriptive analysis. An extension to more than one analysis in the selection stage is conceptually straightforward.

At the start of the clinical trial, suppose we are interested in m hypotheses $H_i : \theta_i \geq 0, i = 1, \dots, m$, where θ_i denotes the treatment effect with the smaller value indicating better treatment effect. These m hypotheses can correspond to treatment effects across different endpoints, treatment groups, populations, or a combination thereof. Let $I_1 = \{1, 2, \dots, m\}$ denote the index set of these m hypotheses. At the end of the adaptive stage, a subset $I_2 \subseteq I_1$ of hypotheses will be selected. The unselected hypotheses will not be tested, so no claim will be made for them. After the selection, further evidence will be collected to test the remaining hypotheses using a group-sequential design, with the potential for an early efficacy claim.

One of the most common selection scenarios is treatment selection, in which the potentially more efficacious treatment(s) will be selected based on pre-specified selection criteria. The hypotheses related to the unselected treatment(s) will no longer be of interest, as typically no patients will be assigned to the unselected treatment group(s). Another common scenario is population selection, where the population(s) demonstrating greater treatment benefits will be selected, and recruitment from the unselected population(s) will be stopped.

Regulatory agencies often require that the subset I_2 be selected according to pre-specified adaptation rules. These selection rules can be quite general and based on primary, secondary, and/or surrogate endpoints. In some cases, however, these rules may not be strictly followed to allow flexibility in the selection process based on the totality of data, ensuring that the most effective treatment and/or the most beneficial patient population can be selected. In all scenarios, it is required that the selection is only based on the Stage-1 data.

Table 1 summarizes the test statistics from two stages. Suppose for $i = 1, \dots, m$, we have the test statistics $X_i^{[1]}$ for hypothesis H_i from the Stage-1 data. If H_i is selected, we will have a sequence of test statistics $\{X_{ij}^{[2]}, j = 1, \dots, s\}$, where s is the number of looks in the group-sequential stage. We assume that the test statistics of $X_i^{[1]}$ and each $X_{ij}^{[2]}$ are standardized with unit variance. For categorical or continuous endpoints based on instantaneous response, the test statistics $X_{ij}^{[2]}$ are derived entirely from the patients enrolled in Stage-2 and therefore independent from $X_i^{[1]}$. For time-to-event endpoints, by using the log-rank test statistics, $X_{ij}^{[2]}$ consists of two components: one based on newly enrolled patients in Stage-2 and the other based on an additional follow-up of patients from Stage-1 (see more details in Section 4.2). The first component is independent of $X_i^{[1]}$, and due to the independent increment property of the log-rank test statistics, the second component is also independent of $X_i^{[1]}$. Therefore, $X_i^{[1]}$ and $[X_{i1}^{[2]}, \dots, X_{is}^{[2]}]$ are independent. Without loss of generality, we assume that $X_i^{[1]}$ follows a normal distribution with unit variance. When H_i is

Table 1. Test statistics in different stages.

Hypothesis	Stage-1 data	Stage-2 (group-sequential stage) data
H_1	$X_1^{[1]}$	$X_{11}^{[2]}, \ldots, X_{1s}^{[2]}$
$\vdots$	$\vdots$	$\vdots$
H_m	$X_m^{[1]}$	$X_{m1}^{[2]}, \ldots, X_{ms}^{[2]}$

selected, $[X_{i1}^{[2]}, \ldots, X_{is}^{[2]}]$ is assumed to follow a multivariate normal distribution with variances equal to one. These assumptions often hold at least asymptotically.

We first define the nominal one-sided p-values

$$p_i^{[1]} = \Phi(X_i^{[1]}), \quad i = 1, \ldots, m,$$
$$p_{ij}^{[2]} = \Phi(X_{ij}^{[2]}), \quad i = 1, \ldots, m, \quad j = 1, \ldots, s,$$

For easy presentation, we only consider one-sided p-values throughout the paper, and let $\Phi(\cdot)$ denote the standard normal distribution function. Note that the above nominal p-values are not used for formal hypothesis testing in Stage-1; rather, they will be used for inference in Stage-2. For the unselected hypothesis H_i, the Stage-2 data may not exist, in this case, $X_{ij}^{[2]}$ are set to $+\infty$ as a convention. Second, we define the weighted statistics

$$Z_{ij} = w_{i1} X_i^{[1]} + w_{i2} X_{ij}^{[2]} i = 1, \ldots, m, \quad j = 1, \ldots, s, \quad (1)$$

with the potentially hypothesis-specific weights satisfying the conditions that $w_{i1} \geq 0$, $w_{i2} \geq 0$ and $w_{i1}^2 + w_{i2}^2 = 1$. In general, the weight functions or algorithms are pre-specified at the start of the study, which can be either fixed or proportional to the planned sample size or expected number of events (i.e., the planned information fraction at each interim) in each stage. For example, suppose the expected number of events for evaluating the hypothesis H_i is E_{i1} in Stage-1 and E_{i2} in Stage-2. The weight functions can be calculated by:

$$w_{i1} = \sqrt{E_{i1}/(E_{i1} + E_{i2})},$$
$$w_{i2} = \sqrt{E_{i2}/(E_{i1} + E_{i2})}.$$

Note that, under the assumption of a single adaptation for hypothesis selection conducted at the end of Stage-1, the weight w_{i2} remains consistent at all interim looks in Stage-2. In contrast, if the design is extended to include multiple adaptive selection stages, the weight functions would need to be updated at each stage accordingly (Sugitani, Bretz, and Maurer 2016).

3.1. Some Intuitive Approaches

One intuitive approach is to apply the closed testing principle in the group-sequential procedure (Marcus, Peritz, and Gabriel 1976; Tang and Geller 1999; Maurer and Bretz 2013). Suppose $\mathcal{M}_1$ and $\mathcal{M}_2$ are closed testing procedures that will be applied to the hypotheses in I_1 and I_2, respectively, where $\mathcal{M}_1$ can be any closed testing procedure, but $\mathcal{M}_2$ may be decided based on the set I_2 of the selected hypotheses (i.e., determined based on the Stage-1 results). Note that $\mathcal{M}_2$ can be completely different from $\mathcal{M}_1$ even when $I_2 = I_1$. For example, $\mathcal{M}_1$ can be a weighted Bonferroni-Holm procedure and $\mathcal{M}_2$ can be a Hochberg procedure. A coherent way to specify $\mathcal{M}_2$ is to use the procedure $\mathcal{M}_1$ to generate a closed testing tree for the hypotheses in I_2 based on the graphical approach (Bretz et al. 2011). In this way, only one closed testing procedure needs to be specified for the study, and it will be automatically adapted to the selection process. $\mathcal{M}_2$ is kept as a general closed testing procedure for maximum flexibility in the following discussions.

For any non-empty subset $G_1 \subseteq I_1$, to test the intersection hypothesis $H_{G_1} = \cap_{i \in G_1} H_i$, one can define the adjusted p-values for Stage-1 and Stage-2 respectively as

$$p^{adj,1}(G_1) = f_{\mathcal{M}_1}\left(G_1, \{p_i^{[1]} : i \in I_1\}\right)$$
$$p_j^{adj,2}(G_1) = f_{\mathcal{M}_2}\left(G_1 \cap I_2, \{p_{ij}^{[2]}, i \in I_2\}\right), \quad j = 1, \ldots, s, \quad (2)$$

where the generic function f computes the adjusted p-value. In general, suppose a closed testing procedure $\mathcal{M}$ is used to test a set of hypotheses $\{H_i : i \in I\}$ based on the corresponding marginal p-values $\{p_i : i \in I\}$, then for any non-empty subset $G \subseteq I$, the procedure will produce an adjusted p-value to test the intersection hypothesis $H_G = \cap_{i \in G} H_i$, which depends on the subset G, the closed testing procedure $\mathcal{M}$ and the set of p-values $\{p_i : i \in I\}$. The adjusted p-value is sometimes called the local p-value as it is used to test the "local" hypothesis H_G (Wright 1992). The adjusted p-values for the commonly used procedures such as Bonferroni, Simes, Holm, Hochberg and Hommel (Holm 1979; Hochberg 1988; Hommel 1988; Shaffer 1995; Sarkar and Chang 1997; Gou and Tamhane 2018) can be calculated using the function `p.adjust` from the R package `stats` (R Core Team 2023). Note that the R function produces adjusted p-values for individual hypotheses, and the adjusted p-value for the intersection hypothesis H_G is the minimum of the individual adjusted p-values over the indexes in the set G. For example, suppose $G_1 = \{1, 2\}$ and $\mathcal{M}_1$ is the standard Bonferroni procedure. Then the adjusted p-value for the intersection hypothesis $H_{\{1,2\}}$ is $p^{adj,1}(G_1) = f_{\mathcal{M}_1}\left(G_1, \{p_1^{[1]}, p_2^{[1]}\}\right) = 2 \times \min(p_1^{[1]}, p_2^{[1]})$. Furthermore, the adjusted p-values for the graphical procedures (Bretz et al. 2011) can be derived using the α-propagation method described therein to construct the full closure tree. Similarly, the adjusted p-values for the gatekeeping procedures can be derived based on the approaches in Dmitrienko and Tamhane (2011) via the corresponding full closure trees. Finally, note that any assumptions that are necessary for the selected testing procedure $\mathcal{M}$ should be evaluated and verified in advance.

With the adjusted p-values defined in Equations (2), the inverse normal combination test statistics for testing H_{G_1} are defined as

$$Z_j^{adj}(G_1) = w_1(G_1)\Phi^{-1}\left\{p^{adj,1}(G_1)\right\}$$
$$+ w_2(G_1)\Phi^{-1}\left\{p_j^{adj,2}(G_1)\right\}, \quad j = 1, \ldots, s, \quad (3)$$

where the weight $w_1(G_1)$ is pre-specified and can depend on $\{w_{i1}, i \in G_1\}$ and $w_2(G_1) = \sqrt{1 - w_1^2(G_1)}$. Based on the closed testing principle (Marcus, Peritz, and Gabriel 1976), to reject H_i for $i \in I_2$ with the family-wise Type I error rate strongly controlled at level α, it is required that for every G_1 such that $i \in G_1 \cap I_2$, there exists at least one $j = 1, \ldots, s$ (could depend on G_1), such that $Z_j^{adj}(G_1) \leq \Phi^{-1}(\alpha)$.

Keen readers may have noticed a limitation in the above statement. The procedure does not account for the group-sequential feature of the design, leading to inflation in the Type I error rate since H_{G_1} is repeatedly tested s times at level α. A quick remedy would be to start "spending" α to the sequence of test statistics $\{Z_j^{adj}(G_1), j = 1,\dots,s\}$ based on their joint distribution. However, the issue with this method is that the adjusted p-values defined in Equations (2) do not necessarily follow the uniform distribution $U(0,1)$ when $\theta_i = 0$ for all $i \in G_1$. In many situations, we are only certain that they are stochastically larger than the uniformly distributed random variable. Consequently, the test statistics $\{Z_j^{adj}(G_1), j = 1,\dots,s\}$ may not follow a joint multivariate normal distribution. Their joint distribution is analytically complex and is highly dependent on the closed testing procedures $\mathcal{M}_1$ and $\mathcal{M}_2$. This is the case even for the simplest scenario when $\mathcal{M}_1$ is the Bonferroni procedure, and only one hypothesis will be selected for Stage-2. Without an easy way to compute the joint distribution, it is difficult to compute the efficacy stopping boundary, making group-sequential testing unreliable.

3.2. The Proposed Approach

This section presents a strategy for multiplicity control in trial design with adaptive selection followed by group-sequential testing. This strategy builds on established concepts in adaptive design and multiplicity control, including inverse normal combination test (Lehmacher and Wassmer 1999), the closed testing principle (Marcus, Peritz, and Gabriel 1976), and the group-sequential p-values (Luo and Quan 2024).

We first define the group-sequential p-values, a stage-wise p-values following the similar idea in Luo and Quan (2024). In the group-sequential testing setting, other types of p-values can be defined based on different orderings of the sample space (see Jennison and Turnbull 1999, pp 179–180). In this study, we adopt the stage-wise ordering version of the p-values (Luo and Quan 2024) because their monotonicity property makes them particularly suitable for the application in multiple testing in sequential trials.

For the unselected hypothesis H_i, the group-sequential p-values for Stage-2 data are set to 1 as no rejection will be made, that is, $\tilde{p}_{ij}^{[2]} = 1$, for $i \notin I_2$ and $j = 1,\dots,s$. For the selected hypothesis H_i, we will define the group-sequential p-values $\{\tilde{p}_{ij}^{[2]}, j = 1,\dots,s\}$, based on $X_i^{[1]}$, $[X_{i1}^{[2]},\dots,X_{is}^{[2]}]$ and a sequence of cumulative error rates $0 \le \alpha_{i1} \le \dots \le \alpha_{is} \le \alpha$. The α-sequence is often determined via some error-spending function based on the information fraction pertaining to the hypothesis, such as Pocock (Pocock 1977) and O'Brien-Fleming (O'Brien and Fleming 1979) methods. However, we keep the sequence general here. Recall that the weighted test statistics defined in Equation (1)

$$Z_{ij} = w_{i1}X_i^{[1]} + w_{i2}X_{ij}^{[2]}, \quad i \in I_2, \quad j = 1,\dots,s,$$

with pre-specified weights w_{i1} and w_{i2}. The group-sequential p-values are defined via the following three steps. An illustrative example and the corresponding code is provided on GitHub at https://github.com/kimihua1995/Adapt_GroupSeq.

Step 1: Compute the critical values $g_{i1},\dots,g_{is}$ recursively such that

$$\begin{aligned}
&P(Z_{i1} \le g_{i1}) = \alpha_{i1}\\
&P(Z_{i1} \le g_{i1} \text{ or } Z_{i2} \le g_{i2}) = \alpha_{i2}\\
&\dots\\
&P(Z_{i1} \le g_{i1} \text{ or } \dots, Z_{iu} \le g_{iu}) = \alpha_{iu}\\
&\dots\\
&P(Z_{i1} \le g_{i1} \text{ or } \dots, Z_{is} \le g_{is}) = \alpha_{is},
\end{aligned}$$

where the probabilities are computed under $\theta_i = 0$. Here we assume that smaller values tend to reject the null hypotheses. This step can be easily accomplished because $\{Z_{ij}, j = 1,\dots,s\}$ follows a multivariate normal distribution with mean zero and covariance matrix Σ_i with the element at j^{th} row and j'^{th} column equal to $w_{i1}^2 + w_{i2}^2\mathrm{Cov}(X_{ij}^{[2]}, X_{ij'}^{[2]}), j, j' = 1,\dots,s$.

Step 2: Let $c_{ij} = (g_{ij} - w_{i1}X_i^{[1]})/w_{i2}$ for $j = 1,\dots,s$ and define the sub-distribution functions given $X_i^{[1]}$ as

$$F_{i1}(x) = P\left\{X_{i1}^{[2]} \le x \middle| X_i^{[1]}\right\} = P\left\{X_{i1}^{[2]} \le x\right\}$$

by the independence of $X_{ij}^{[2]}$ and $X_i^{[1]}$

$$\begin{aligned}
F_{i2}(x) &= P\left\{X_{i2}^{[2]} \le x, X_{i1}^{[2]} > c_{i1} \middle| X_i^{[1]}\right\}\\
F_{i3}(x) &= P\left\{X_{i3}^{[2]} \le x, X_{i2}^{[2]} > c_{i2}, X_{i1}^{[2]} > c_{i1} \middle| X_i^{[1]}\right\}\\
&\dots\\
F_{is}(x) &= P\left\{X_{is}^{[2]} \le x, X_{i,s-1}^{[2]} > c_{i,s-1},\dots, X_{i1}^{[2]} > c_{i1} \middle| X_i^{[1]}\right\}.
\end{aligned}$$

and compute the cumulative errors $\alpha_{iu}^c = \sum_{j=1}^u F_{ij}(c_{ij})$, $u = 1,\dots,s$. Here, the probabilities are also computed under $\theta_i = 0$ and the functions are easily computable as the distribution of $\{X_{ij}^{[2]}, j = 1,\dots,s\}$ is a multivariate normal distribution with mean zero and covariance matrix Ω_i with the element at the j^{th} row and j'^{th} column equal to $\mathrm{Cov}(X_{ij}^{[2]}, X_{ij'}^{[2]}), j, j' = 1,\dots,s$. Note that the Stage-1 test statistics $X_i^{[1]}$ only appears in the critical values c_{ij}.

Step 3: Finally, the group-sequential p-values $\{\tilde{p}_{ij}^{[2]}, j = 1,\dots,s\}$ are

$$\begin{aligned}
\tilde{p}_{i1}^{[2]} &= I(X_{i1}^{[2]} \le c_{i1})F_{i1}(X_{i1}^{[2]}) + I(X_{i1}^{[2]} > c_{i1})\\
\tilde{p}_{i2}^{[2]} &= I(X_{i1}^{[2]} \le c_{i1})F_{i1}(X_{i1}^{[2]}) + I(X_{i1}^{[2]} > c_{i1}, X_{i2}^{[2]}\\
&\quad \le c_{i2})\left\{\alpha_{i1}^c + F_{i2}(X_{i2}^{[2]})\right\} + I(X_{i1}^{[2]} > c_{i1}, X_{i2}^{[2]} > c_{i2})\\
&\quad \dots\\
\tilde{p}_{i,s-1}^{[2]} &= I(X_{i1}^{[2]} \le c_{i1})F_{i1}(X_{i1}^{[2]}) + I(X_{i1}^{[2]} > c_{i1}, X_{i2}^{[2]}\\
&\quad \le c_{i2})\left\{\alpha_{i1}^c + F_{i2}(X_{i2}^{[2]})\right\}\\
&\quad + \dots + I(X_{i1}^{[2]} > c_{i1},\dots, X_{i,s-2}^{[2]} > c_{i,s-2}, X_{i,s-1}^{[2]}\\
&\quad \le c_{i,s-1})\left\{\alpha_{i,s-2}^c + F_{i,s-1}(X_{i,s-1}^{[2]})\right\}\\
&\quad + I(X_{i1}^{[2]} > c_{i1},\dots, X_{i,s-1}^{[2]} > c_{i,s-1})\\
\tilde{p}_{is}^{[2]} &= I(X_{i1}^{[2]} \le c_{i1})F_{i1}(X_{i1}^{[2]}) + I(X_{i1}^{[2]} > c_{i1}, X_{i2}^{[2]}
\end{aligned}$$

$$\leq c_{i2})\left\{\alpha_{i1}^c + F_{i2}(X_{i2}^{[2]})\right\} + \ldots + I(X_{i1}^{[2]} > c_{i1}, \ldots, X_{i,s-2}^{[2]} > c_{i,s-2}, X_{i,s-1}^{[2]} \leq c_{i,s-1})\left\{\alpha_{i,s-2}^c + F_{i,s-1}(X_{i,s-1}^{[2]})\right\} + I(X_{i1}^{[2]} > c_{i1}, \ldots, X_{i,s-1}^{[2]} > c_{i,s-1})\left\{\alpha_{i,s-1}^c + F_{is}(X_{is}^{[2]})\right\}.$$

The group-sequential p-values defined above are similar to the group-sequential p-values proposed in Luo and Quan (2024), except that the critical values are derived based on the Stage-1 test statistics $X_i^{[1]}$.

The basic idea of defining group-sequential p-values is that, at each interim look (excluding the final analysis), if the test statistic crosses the efficacy boundary, the p-value is set to its actual value. Otherwise, the p-value is set to one as we should not make a claim. The group-sequential p-values have the following properties (Luo and Quan 2024).

(I) $\tilde{p}_{i1}^{[2]} \geq \tilde{p}_{i2}^{[2]} \geq \cdots \geq \tilde{p}_{is}^{[2]}$. This is obvious from their definition.
(II) $\tilde{p}_{is}^{[2]} \sim U(0, 1)$ given $\theta_i = 0$. Therefore, $\tilde{p}_{is}^{[2]}$ is a proper p-value. The proof can be established following the same steps detailed in Luo and Quan (2024).

With the group-sequential p-values $\{\tilde{p}_{ij}^{[2]}, j = 1, \ldots, s\}$, we can apply the closed testing principle (Marcus, Peritz, and Gabriel 1976) to conduct hypothesis testing. Suppose that the closed testing procedures $\mathcal{M}_1$ and $\mathcal{M}_2$ are used respectively for Stage-1 with the original set I_1 of the hypotheses and for Stage-2 with the selected set I_2 of the hypotheses. To test the intersection hypotheses $H_{G_1} = \cap_{i \in G_1} H_i$ for any non-empty subset $G_1 \subseteq I_1$, we can use the following inverse normal combination test statistics

$$\tilde{Z}_j^{adj}(G_1) = w_1(G_1)\Phi^{-1}\left\{p^{adj,1}(G_1)\right\} + w_2(G_1)\Phi^{-1}\left\{\tilde{p}_j^{adj,2}(G_1)\right\}, \quad j = 1, \ldots, s, \tag{4}$$

where $p^{adj,1}(G_1)$ is defined in Equations (2) and

$$\tilde{p}_j^{adj,2}(G_1) = f_{\mathcal{M}_2}\Big(G_1 \cap I_2, \{\tilde{p}_{ij}^{[2]}, i \in I_2\}\Big), \quad j = 1, \ldots, s. \tag{5}$$

Based on the closed testing principle (Marcus, Peritz, and Gabriel 1976), to reject H_i for $i \in I_2$ with the strongly controlled family-wise Type I error rate at level α, we will require that for every G_1 such that $i \in G_1 \cap I_2$, there exists a $j \in \{1, \ldots, s\}$ (which could depend on G_1), such that $\tilde{Z}_j^{adj}(G_1) \leq \Phi^{-1}(\alpha)$.

The only difference between the test statistics $\tilde{Z}_j^{adj}(G_1)$ and the test statistics $Z_j^{adj}(G_1)$ defined in Equation (3) is that $\tilde{Z}_j^{adj}(G_1)$ replaces the adjusted p-value $p_j^{adj,2}(G_1)$ defined in Equations (2) with the adjusted p-value $\tilde{p}_j^{adj,2}(G_1)$ defined based on the group-sequential p-values $\{\tilde{p}_{ij}^{[2]} : i \in I_2\}$ in Equation (5).

Additionally, this difference leads to a valid procedure that strongly controls the family-wise Type I error rate for the study with adaptive selection followed by group-sequential testing. Because the group-sequential p-values are not increasing, that is, $\tilde{p}_{i1}^{[2]} \geq \tilde{p}_{i2}^{[2]} \geq \cdots \geq \tilde{p}_{is}^{[2]}$ for all $i \in I_2$, intuitively the adjusted p-values should not increase, that is, $\tilde{p}_1^{adj,2}(G_1) \geq \tilde{p}_2^{adj,2}(G_1) \geq \cdots \geq \tilde{p}_s^{adj,2}(G_1)$. As such, $\tilde{Z}_1^{adj}(G_1) \geq \tilde{Z}_2^{adj}(G_1) \geq \cdots \geq \tilde{Z}_s^{adj}(G_1)$. Therefore, an early rejection of H_{G_1} will imply a rejection of H_{G_1} at any later analyses. Furthermore, for the selected H_i, $\tilde{p}_{is}$ is a proper p-value because $\tilde{p}_{is}^{[2]} \sim U(0, 1)$ given $\theta_i = 0$, leading to strong control of the family-wise Type I error rate at or below the level α by implementing the inverse normal combination test and the closed testing principle.

The missing piece for the above argument is that we require the closed testing procedure $\mathcal{M}_2$ to satisfy a monotone property. Specifically, suppose that a closed testing procedure $\mathcal{M}$ is used to test hypotheses $H_1, \ldots, H_m$. Suppose we have a sequence of p-values $\{p'_1, \ldots, p'_m\}$ such that it is component-wise not smaller than another sequence of p-values $\{p_1, \ldots, p_m\}$, that is, $p'_i \geq p_i$ for $i = 1, \ldots, m$, we say the procedure $\mathcal{M}$ satisfies the monotone property if, for any $G \subseteq \{1, \ldots, m\}$,

$$f_{\mathcal{M}}\Big(G, \{p'_1, \ldots, p'_m\}\Big) \geq f_{\mathcal{M}}\Big(G, \{p_1, \ldots, p_m\}\Big). \tag{6}$$

In other words, if $H_G = \cap_{i \in G} H_i$ can be rejected using larger p-values $\{p'_1, \ldots, p'_m\}$, it must be rejected with smaller p-values $\{p_1, \ldots, p_m\}$. An equivalent statement to (6) is to say that the hypotheses rejected based on p-values $\{p'_1, \ldots, p'_m\}$ must also be rejected based on p-values $\{p_1, \ldots, p_m\}$. The graphical procedure (Bretz et al. 2009), Hommel procedure (Hommel 1988), and Hochberg procedure (Hochberg 1988) have been shown to satisfy this property (Luo and Quan 2024). The appendix shows that the commonly used weighted procedures have this property.

4. Simulation Study

This simulation study aims to evaluate the finite-sample properties of the proposed testing method involving multiple treatments, subgroups, and endpoints, and to characterize the operating characteristics of the design. Building on the aforementioned motivating example, we simulate a two-stage adaptive oncology clinical trial evaluating two experimental treatments (T_1 and T_2) against the standard of care (T_0) in both the biomarker positive (BM+) and overall populations. The efficacy evaluation is based on two primary endpoints: progression-free survival (PFS) and overall survival (OS). Treatment and population selection occur at the end of the first stage (IA1), based on a short-term endpoint, the objective response (ORR). All hypotheses are listed in Table 2. The relevant R codes of the simulation studies are available on Github at https://github.com/kimihua1995/Adapt_GroupSeq.

4.1. Design Settings

Each study participant has three outcomes: 1) the binary outcome ORR (Y^{ORR}), 2) the time-to-event outcome PFS (Y^{PFS}), and 3) the time-to-event outcome OS (Y^{OS}). We use the following procedure to generate the PFS time, OS time, and ORR to impose a correlation structure between these three outcomes. First, as detailed in Table 3, let $l = 1, \ldots, 6$ as the index of each biomarker and treatment subgroup, and $k = 1, \ldots, N_l$ index the individual participant in each subgroup, where N_l is the sample

Table 2. List of hypotheses in the motivation design.

H_1 :	T_2 vs. T_0 in BM+ for PFS	H_5 :	T_2 vs. T_0 in BM+ for OS
H_2 :	T_1 vs. T_0 in BM+ for PFS	H_6 :	T_1 vs. T_0 in BM+ for OS
H_3 :	T_2 vs. T_0 in the overall population for PFS	H_7 :	T_2 vs. T_0 in the overall population for OS
H_4 :	T_1 vs. T_0 in the overall population for PFS	H_8 :	T_1 vs. T_0 in the overall population for OS

Table 3. Index of subgroup, median response rate of ORR (q_l), hazard rates of the exponential distribution for OS and PFS and their median times (in years) in each treatment and biomarker subgroup.

	BM-			BM+		
Index	T_0 $l = 1$	T_1 $l = 2$	T_2 $l = 3$	T_0 $l = 4$	T_1 $l = 5$	T_2 $l = 6$
q_l	0.20	0.25	0.30	0.20	0.35	0.50
λ_l^{PFS}	log(2)	log(2)/1.1	log(2)/1.2	log(2)	log(2)/1.6	log(2)/1.8
λ_l^{OS}	log(2)/2.0	log(2)/2.2	log(2)/2.4	log(2)/2.0	log(2)/3.4	log(2)/3.8
m^{PFS}	1.0	1.1	1.2	1.0	1.6	1.8
m^{OS}	2.0	2.2	2.4	2.0	3.4	3.8

size in each subgroup, we generate two independent exponential random variables:

X_{k1l}: with hazard rate $\lambda_l^{PFS} - \lambda_l^{OS}$,
X_{k2l}: with hazard rate λ_l^{OS}.

Second, we set $T_{kl}^{OS} = X_{k2l}$, which follows an exponential distribution with hazard rate λ_l^{OS}, and $T_{kl}^{PFS} = \min\{X_{k1l}, X_{k2l}\}$, which by construction follows an exponential distribution with hazard rate λ_l^{PFS}. By construction, $T_{kl}^{OS} \geq T_{kl}^{PFS}$ and are positively correlated with each other. Third, for each participant k in subgroup l, we generate the binary outcome of ORR with a following logistic relationship between PFS, OS, and ORR:

$$\text{logit}(Y_{kl}^{ORR}) = p_l + \beta_1 \log(T_{kl}^{PFS}) + \beta_2 \log(T_{kl}^{OS}).$$

Here, the positive values of β_1 and β_2 allow a positive correlation between PFS, OS, and ORR. We derive the values of p_l based on a fixed median response rate of q_l according to the following relationship with median time of PFS and OS in each treatment and biomarker subgroup:

$$\begin{aligned} q_l &= \frac{1}{1 + \exp\{-p_l - \beta_1 \log(\text{median}[T_{kl}^{PFS}]) - \beta_2 \log(\text{median}[T_{kl}^{OS}])\}} \\ &= \frac{1}{1+\exp\{-p_l - \beta_1 \log(\log(2)/\lambda_l^{PFS}) - \beta_2 \log(\log(2)/\lambda_l^{OS})\}}. \end{aligned}$$

The values of q_l, λ_l^{PFS}, and λ_l^{OS} for each subgroup are listed in Table 3.

The adaptive selection based on ORR at the end of Stage-1 follows the selection criteria as described in Section 2. In addition, since the design timeline is based on comparisons between T_2 and T_0 in the BM+ population, each interim analysis is conducted when a specific number of PFS or OS events have been observed. Four analyses are conducted as

IA1 (t_1): Achieve 25% of the target number of PFS events.
IA2 (t_2): Achieve 75% of the target number of PFS events.
IA3 (t_3): Achieve 100% of the target number of PFS events.
FA (t_4): Achieve 100% of the target number of OS events.

Note that since 100% of the target number of PFS events will be achieved at IA3, we will use the same data cut of IA3 as the FA for the PFS analyses, that is, H_1 to H_4. The patients will be kept following-up for the OS analyses.

According to Table 3, the hazard ratios comparing T_2 and T_0 in the BM+ population for PFS and OS are 0.56 and 0.53, respectively. Assume the accrual duration is three years with a minimum follow-up time of one year, based on the Schoenfeld formula (Schoenfeld 1981), 166 PFS events and 160 OS events are required to both achieve 95% power in a fixed design with one-sided alpha of 0.0167 and 0.0083, respectively. Given these parameters, the study plans to recruit 200 patients in each biomarker and treatment subgroup, resulting in a maximum of 1,200 patients in total. For simplicity, we assume the patient recruitment rate is a constant for the entire three-year recruitment period. With these assumptions, the four analyses will occur roughly at month $t_1 = 15$, $t_2 = 28$, $t_3 = 34$, and $t_4 = 47$ (derived through simulation).

We assume the loss-to-follow-up time L follows an exponential distribution with a hazard rate of 0.1, independent of the treatment and biomarker subgroup. For the interim analysis conducted at time t_j ($j = 1, 2, 3, 4$), the censoring time for each patient is defined as $C = \min(L, t_j - E)$, where $E \sim \text{Unif}(0, 3)$ denotes the enrollment time. Consequently, the time-to-PFS (Y^{PFS}) and time-to-OS (Y^{OS}) are subject to the censoring time C, such that $Y^{PFS} = \min(T^{PFS}, C)$ and $Y^{OS} = \min(T^{OS}, C)$.

For each local test within the closed testing procedures, we use weighted Simes test (Hochberg and Liberman 1994) and weighted Bonferroni test (Holm 1979) for both $\mathcal{M}_1$ and $\mathcal{M}_2$ in separate scenarios to test the intersection hypotheses using the nominal p-values in Stage-1 and group-sequential p-values in Stage-2. By defining an initial weight of $W(I_1) = [W_1, \dots, W_8] = [2/3, 0, 0, 0, 1/3, 0, 0, 0]$ and employing a transition matrix as described in Figure 3 for hypotheses H_1 to H_8, we utilize the weighting strategy as proposed in Bretz et al. (2011) to update the weights of weighted Simes and Bonferroni tests for each intersection hypothesis.

Following Maurer and Bretz (2013), we assign the $\alpha-$sequence $\{\alpha_{i1}, \alpha_{i2}, \alpha_{i3}\}$ using the power spending function $\alpha_{ij} = \alpha \times \gamma_{ij}^{\rho}$, where γ_{ij} denotes the information fraction based on the planned number of events for hypothesis H_i at j^{th} interim in Stage-2, and $\alpha = 0.025$ is the overall one-sided Type I error rate. We set $\rho = 3$ to obtain similar to the

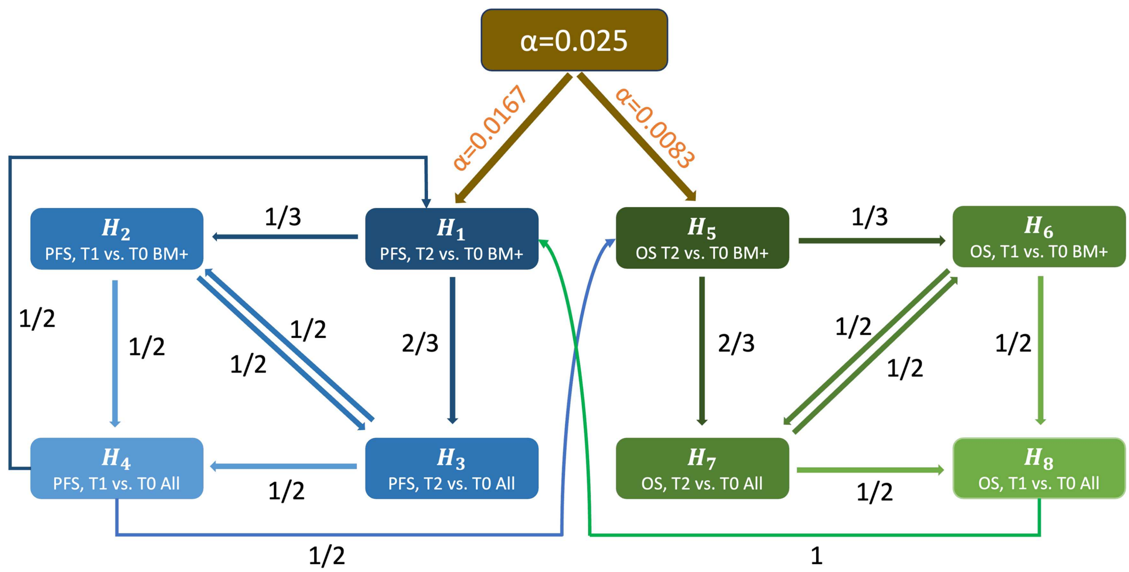

Figure 3. Graphical procedure for testing PFS and OS in the closed testing procedure.

O'Brien-Fleming boundaries (Jennison and Turnbull 1999). For PFS-related hypotheses H_1 to H_4, as 100% of the target number of PFS events will be observed at IA3 and the same data cut of IA3 will be used at FA for the PFS analyses, we assign the $\alpha-$ sequence as $[\alpha_{i1}, \alpha_{i2}, \alpha_{i3}] = [0.75^3, 1, 1.01] \times 0.025 \times W_i$ for $i = 1, \dots, 4$. For OS-related hypotheses H_5 to H_8, we assign the $\alpha-$ sequence as $[\alpha_{i1}, \alpha_{i2}, \alpha_{i3}] = [0.1, 0.25, 1] \times 0.025 \times W_i$ for $i = 5, \dots, 8$. The planned number of events for each hypothesis are derived using the simulation method (see details at https://github.com/kimihua1995/Adapt_GroupSeq). Since the information fractions are similar within PFS hypotheses (H_1 to H_4) and OS hypotheses (H_5 to H_8), we assign the same $\alpha-$ sequence among all PFS and OS hypotheses, respectively.

Additionally, we compare the proposed design with an existing graphical testing procedures for adaptive group-sequential clinical trials (Sugitani, Bretz, and Maurer 2016). To ensure a fair comparison, we modify the existing approach to accommodate the settings in our simulation study. In the comparator design, the dropped hypotheses after Stage-1 will no longer be tested in Stage-2, and the test statistics are defined following Equation (1) as $Z_{ij} = w_{i1}X_i^{[1]} + w_{i2}X_{ij}^{[2]}$, where $i \in I_2$ and $j = 1, 2, 3$. In Stage-2, the comparator design will exactly follow the group sequential testing procedure using weighted Bonferroni test as proposed by Maurer and Bretz (2013), where the graph used in the graphical procedure will be a subgraph of Figure 3 by only keeping the nodes and edges corresponding to the remaining hypotheses in I_2. For example, if $I_2 = \{H_1, H_2, H_5, H_6\}$, the resulting subgroup is shown in Figure 4.

4.2. Test Statistics

The test statistics $X_i^{[1]}$ from Stage-1 and a sequence test statistics $\{X_{ij}^{[2]}, j = 1, 2, 3\}$, from Stage-2 are determined based on each hypothesis H_i ($i = 1, \dots, 8$). At the end of Stage-1, we denote $L_i^{[1]}(t_1)$ as the log-rank score for testing H_i at t_1, with approximate variance of $D_i^{[1]}(t_1)/4$, where $D_i^{[1]}(t)$ is the total number of events observed by time t about testing hypothesis H_i for patients recruited in Stage-1. The test statistics $X_i^{[1]}$ is formulated as $2L_i^{[1]}(t_1)/\sqrt{D_i^{[1]}(t_1)}$.

In Stage-2, the group-sequential test statistic $X_{ij}^{[2]}$ is defined by three parts based on the interim analysis at time t_j, for $j = 2, 3, 4$. First, we conduct a log-rank test for patients enrolled before t_1, we denote $L_i^{[1]}(t_j)$ as the log-rank score with a variance of $D_i^{[1]}(t_j)/4$. Second, we conduct an additional log-rank test by restricting the patients enrolled after t_1, we denote $L_i^{[2]}(t_j)$ as the log-rank score with variance of $D_i^{[2]}(t_j)/4$, where $D_i^{[2]}(t)$ is the number of events observed by time t to test hypothesis H_i for patients recruited in Stage-2. Third, we subtract $L_i^{[1]}(t_1)$ from $L_i^{[1]}(t_j)$ to ensure that $X_{ij}^{[2]}$ is conditionally independent given the Stage-1 data. Consequently,

$$X_{ij}^{[2]} = \frac{2\left\{L_i^{[2]}(t_j) + L_i^{[1]}(t_j) - L_i^{[1]}(t_1)\right\}}{\sqrt{D_i^{[2]}(t_j) + D_i^{[1]}(t_j) - D_i^{[1]}(t_1)}}.$$

Based on the independent increments property of the log-rank test statistics, we can show that the variance of $L_i^{[2]}(t_j) + L_i^{[1]}(t_j) - L_i^{[1]}(t_1)$ is equal to $\left\{D_i^{[2]}(t_j) + D_i^{[1]}(t_j) - D_i^{[1]}(t_1)\right\}/4$, and $X_{ij}^{[2]}$ is independent with $X_i^{[1]}$.

For $j < j'$, the covariance between the numerators of $X_{ij}^{[2]}$ and $X_{ij'}^{[2]}$ is derived as follows:

$$\begin{aligned}
&\mathrm{cov}(L_i^{[2]}(t_j) + L_i^{[1]}(t_j) - L_i^{[1]}(t_1), L_i^{[2]}(t_{j'}) + L_i^{[1]}(t_{j'}) - L_i^{[1]}(t_1)) \\
&= \mathrm{cov}(L_i^{[2]}(t_j), L_i^{[2]}(t_{j'})) + \mathrm{cov}(L_i^{[1]}(t_j), L_i^{[1]}(t_{j'})) \\
&- \mathrm{cov}(L_i^{[1]}(t_j), L_i^{[1]}(t_1)) \\
&- \mathrm{cov}(L_i^{[1]}(t_1), L_i^{[1]}(t_{j'})) + \mathrm{var}(L_i^{[1]}(t_1)) \\
&= D_i^{[2]}(t_j) + D_i^{[1]}(t_j) - D_i^{[1]}(t_1) - D_i^{[1]}(t_1) + D_i^{[1]}(t_1) \\
&= D_i^{[2]}(t_j) + D_i^{[1]}(t_j) - D_i^{[1]}(t_1) \\
&= \mathrm{var}(L_i^{[2]}(t_j) + L_i^{[1]}(t_j) - L_i^{[1]}(t_1))
\end{aligned}$$

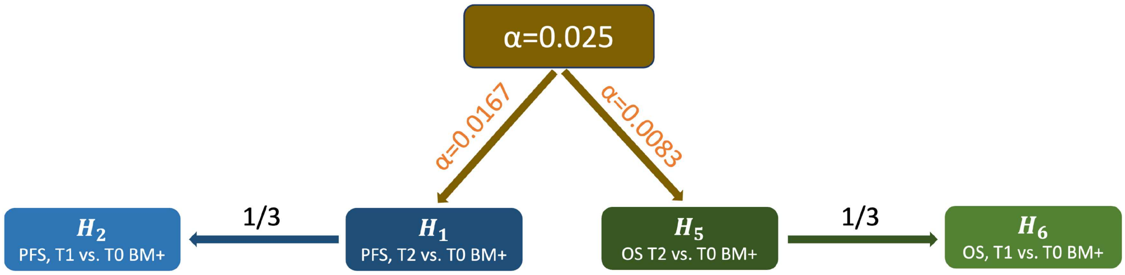

Figure 4. Example of a subgraph of Figure 2 by only keeping H_1, H_2, H_5, H_6 in stage-2 in the graphical procedure in the comparator design.

Table 4. Values of λ_l^{PFS} and λ_l^{OS} in each simulation scenario.

		BM-			BM+		
Scenario		T_0	T_1	T_2	T_0	T_1	T_2
S1	λ_l^{PFS}	log(2)	log(2)/1.1	log(2)/1.2	log(2)	log(2)/1.6	log(2)/1.8
	λ_l^{OS}	log(2)/2.0	log(2)/2.2	log(2)/2.4	log(2)/2.0	log(2)/3.4	log(2)/3.8
S2	λ_l^{PFS}	log(2)	log(2)/1.1	log(2)/1.2	log(2)	log(2)/1.6	log(2)
(null H_1 & H_5)	λ_l^{OS}	log(2)/2.0	log(2)/2.2	log(2)/2.4	log(2)/2.0	log(2)/3.4	log(2)/2.0
S3	λ_l^{PFS}	log(2)	log(2)/1.1	log(2)/1.2	log(2)	log(2)	log(2)/1.8
(null H_2 & H_6)	λ_l^{OS}	log(2)/2.0	log(2)/2.2	log(2)/2.4	log(2)/2.0	log(2)/2.0	log(2)/3.8
S4	λ_l^{PFS}	log(2)	log(2)/1.1	log(2)/1.2	log(2)	log(2)	log(2)
(null H_1 & H_2 & H_5 & H_6)	λ_l^{OS}	log(2)/2.0	log(2)/2.2	log(2)/2.4	log(2)/2.0	log(2)/2.0	log(2)/2.0
S5	λ_l^{PFS}	log(2)	log(2)/1.1	log(2)	log(2)	log(2)/1.6	log(2)
(null H_1 & H_3 & H_5 & H_7)	λ_l^{OS}	log(2)/2.0	log(2)/2.2	log(2)/2.0	log(2)/2.0	log(2)/3.4	log(2)/2.0
S6	λ_l^{PFS}	log(2)	log(2)	log(2)/1.2	log(2)	log(2)	log(2)/1.8
(null H_2 & H_4 & H_6 & H_8)	λ_l^{OS}	log(2)/2.0	log(2)/2.0	log(2)/2.4	log(2)/2.0	log(2)/2.0	log(2)/3.8
S7	λ_l^{PFS}	log(2)	log(2)	log(2)	log(2)	log(2)	log(2)
(null H_1 to H_8)	λ_l^{OS}	log(2)/2.0	log(2)/2.0	log(2)/2.0	log(2)/2.0	log(2)/2.0	log(2)/2.0

Table 5. Proportion of each enrichment selection at IA1.

Scenario	No enrichment	Drop T_1	Drop BM-	Drop T_1 and BM-	Stop all
S1	71.08%	24.32%	1.12%	0.88%	2.60%
S2	69.44%	24.38%	1.44%	0.96%	3.78%
S3	71.46%	23.94%	1.10%	0.90%	2.60%
S4	69.62%	24.20%	1.46%	0.94%	3.78%
S5	69.44%	24.38%	1.44%	0.96%	3.78%
S6	71.46%	23.94%	1.10%	0.90%	2.60%
S7	69.62%	24.20%	1.46%	0.94%	3.78%

Therefore, the correlation between $X_{ij}^{[2]}$ and $X_{ij'}^{[2]}$ is

$$\sqrt{\frac{D_i^{[2]}(t_j) + D_i^{[1]}(t_j) - D_i^{[1]}(t_1)}{D_i^{[2]}(t_{j'}) + D_i^{[1]}(t_{j'}) - D_i^{[1]}(t_1)}}$$

For the weighted test statistics $Z_{ij} = w_{i1}X_i^{[1]} + w_{i2}X_{ij}^{[2]}$, $\mathrm{corr}(Z_{ij}, Z_{ij'}) = w_{i1}^2 + w_{i2}^2 \times \mathrm{corr}(X_{ij}^{[2]}, X_{ij'}^{[2]})$. For simplicity, we assume fixed weights $w_{i1} = \sqrt{1/4}$ and $w_{i2} = \sqrt{3/4}$ throughout the simulation study.

4.3. Results

In the simulation, we consider seven scenarios based on different hypothesis settings for PFS and OS, which are summarized in Table 4. Three methods are compared within each scenario: (1) the proposed design using weighted Simes test for the closed testing procedure; (2) the proposed design using weighted Bonferroni test for the closed testing procedure; and (3) the comparator design in Sugitani, Bretz, and Maurer (2016) as described in Section 4.1. The simulation results are evaluated under 5,000 Monte Carlo replications, the Monte Carlo standard error of the estimated Type I error is approximately 0.0022.

The results are summarized in Tables 5 to 7. Table 5 shows the proportion of each enrichment condition at IA1, where these proportions remain unaffected by changes in hazard rates, λ_l^{PFS} and λ_l^{OS}, from different scenarios. With fixed values of q_l, approximately 70% of simulations result in no enrichment, around 24% lead to dropping T_1 at IA1, about 1.4% drop the BM- group at IA1, 1% drop both T_1 and BM-, and roughly 3% stop the trial at IA1.

Tables 6 and 7 present the unadjusted and adjusted cumulative rejection probabilities at FA. The adjusted cumulative rejections are calculated by excluding simulations where the corresponding hypotheses were dropped at IA1. Each column of H_i in the tables displays the rejection rates for each single hypothesis across different methods and scenarios, while the last column of H^* presents the maximum Type I error rate in each method and scenario.

Table 6. Unadjusted cumulative rejection probability at FA, comparing three methods: (1) S: proposed design using weighted Simes test, (2) B: proposed design using weighted Bonferroni test, and (3) G: comparator design in Sugitani, Bretz, and Maurer (2016).

Scenario	Methods	H_1	H_2	H_3	H_4	H_5	H_6	H_7	H_8	H^*
S_1	S	92.6%	57.6%	85.2%	49.1%	93.1%	60.9%	84.9%	51.5%	0.0%
	B	90.4%	52.5%	80.6%	43.7%	88.2%	53.3%	77.6%	44.7%	0.0%
	G	92.3%	56.4%	83.5%	46.2%	86.3%	49.6%	73.8%	38.8%	0.0%
S_2	S	1.5%	1.1%	1.0%	0.8%	0.8%	0.6%	0.5%	0.5%	2.0%
	B	1.2%	0.8%	0.7%	0.6%	0.6%	0.4%	0.4%	0.4%	1.6%
	G	1.6%	1.1%	1.0%	0.9%	0.7%	0.6%	0.4%	0.4%	2.2%
S_3	S	90.3%	0.9%	79.2%	1.1%	85.4%	0.5%	72.9%	0.8%	1.3%
	B	88.9%	0.6%	75.9%	0.8%	76.8%	0.3%	60.5%	0.6%	0.7%
	G	92.0%	1.0%	82.1%	1.4%	84.3%	0.2%	71.2%	0.6%	1.1%
S_4	S	1.3%	0.1%	0.7%	0.1%	0.5%	0.0%	0.2%	0.0%	1.7%
	B	1.1%	0.1%	0.5%	0.0%	0.4%	0.0%	0.1%	0.0%	1.4%
	G	1.6%	0.1%	1.0%	0.1%	0.6%	0.0%	0.4%	0.0%	2.2%
S_5	S	1.5%	1.0%	0.3%	0.6%	0.6%	0.4%	0.2%	0.4%	1.9%
	B	1.2%	0.8%	0.2%	0.5%	0.4%	0.3%	0.2%	0.2%	1.5%
	G	1.6%	1.1%	0.4%	0.8%	0.6%	0.5%	0.2%	0.3%	2.2%
S_6	S	90.2%	0.8%	79.0%	0.4%	84.5%	0.4%	71.9%	0.3%	1.3%
	B	88.9%	0.5%	75.9%	0.2%	76.2%	0.3%	59.7%	0.2%	0.8%
	G	92.0%	1.0%	82.1%	0.5%	84.2%	0.2%	71.2%	0.2%	1.4%
S_7	S	1.3%	0.1%	0.3%	0.0%	0.5%	0.0%	0.1%	0.0%	1.7%
	B	1.1%	0.1%	0.2%	0.0%	0.3%	0.0%	0.1%	0.0%	1.4%
	G	1.6%	0.1%	0.3%	0.1%	0.6%	0.0%	0.1%	0.0%	2.2%

H^*: maximum Type I error rate under each scenario. For example, under S_2, $H*$ denotes H_1 or H_5.

Table 7. Adjusted cumulative rejection probability at FA, conditional on the corresponding hypothesis that did not stop at IA1, comparing three methods: (1) S: proposed design using weighted Simes test, (2) B: proposed design using weighted Bonferroni test, and (3) G: comparator design in Sugitani, Bretz, and Maurer (2016).

Scenario	Methods	H_1	H_2	H_3	H_4	H_5	H_6	H_7	H_8	H^*
S_1	S	92.5%	77.1%	86.9%	66.6%	93.0%	61.4%	85.7%	52.0%	0.0%
	B	90.4%	70.2%	82.2%	59.3%	88.2%	53.7%	78.2%	45.1%	0.0%
	G	92.3%	75.4%	85.2%	62.7%	86.3%	50.1%	74.5%	39.1%	0.0%
S_2	S	1.5%	1.4%	1.0%	1.1%	0.8%	0.6%	0.5%	0.5%	2.0%
	B	1.2%	1.1%	0.7%	0.8%	0.6%	0.4%	0.4%	0.4%	1.6%
	G	1.6%	1.5%	1.0%	1.2%	0.7%	0.6%	0.4%	0.4%	2.2%
S_3	S	90.2%	1.2%	80.8%	1.4%	85.3%	0.5%	73.6%	0.8%	1.3%
	B	88.9%	0.7%	77.5%	1.1%	76.8%	0.3%	61.1%	0.6%	0.7%
	G	91.9%	1.4%	83.8%	1.8%	84.2%	0.2%	71.9%	0.6%	1.1%
S_4	S	1.3%	0.1%	0.7%	0.1%	0.5%	0.0%	0.2%	0.0%	1.8%
	B	1.1%	0.1%	0.6%	0.1%	0.4%	0.0%	0.1%	0.0%	1.4%
	G	1.6%	0.2%	1.0%	0.1%	0.6%	0.0%	0.4%	0.0%	2.2%
S_5	S	1.5%	1.4%	0.3%	0.8%	0.6%	0.4%	0.2%	0.4%	1.9%
	B	1.2%	1.0%	0.2%	0.7%	0.4%	0.3%	0.2%	0.2%	1.5%
	G	1.6%	1.5%	0.4%	1.1%	0.6%	0.5%	0.2%	0.3%	2.2%
S_6	S	90.1%	1.0%	80.6%	0.5%	84.4%	0.4%	72.6%	0.3%	1.3%
	B	88.9%	0.6%	77.4%	0.3%	76.2%	0.3%	60.2%	0.2%	0.8%
	G	91.9%	1.3%	83.8%	0.7%	84.2%	0.2%	71.8%	0.2%	1.4%
S_7	S	1.3%	0.1%	0.3%	0.1%	0.5%	0.0%	0.1%	0.0%	1.7%
	B	1.1%	0.1%	0.2%	0.0%	0.3%	0.0%	0.1%	0.0%	1.4%
	G	1.6%	0.2%	0.3%	0.1%	0.6%	0.0%	0.1%	0.0%	2.2%

H^*: maximum Type I error rate under each scenario. For example, under S_2, $H*$ denotes H_1 or H_5.

Based on these findings, three key observations can be made. First, the proposed group-sequential testing design, under both testing methods, demonstrates sufficient power while maintaining well-controlled Type I error in an adaptive setting with multiple treatments, subgroups, and endpoints. Second, within the proposed design, the weighted Simes test consistently yields higher power and a higher maximum Type I error rate than the weighted Bonferroni test across all scenarios and hypotheses. Third, compared with the comparator design using weighted Bonferroni test, the proposed design using the weighted Simes test demonstrates higher power for all OS-related hypotheses (H_4 to H_8) and a lower maximum Type I error rate across all scenarios. For the PFS-related hypotheses (H_1 to H_4), the rejection probability of the proposed design is slightly higher in Scenario 1, slightly lower in Scenarios 3 and 6, and comparable in the remaining scenarios than the comparator design. In contrast, the proposed design using the weighted Bonferroni test is more conservative, exhibiting lower power but also lower maximum Type I error rates than the comparator design.

In conclusion, under the setting of the motivation design, the proposed methods using the weighted Simes test demonstrate higher power of testing the OS-related hypotheses and similar performance of testing the PFS-related hypotheses to the comparator design, while the proposed method with the weighted Bonferroni test trades power for stronger control of the maximum Type I error rate.

5. Discussion

This paper proposes a general framework for addressing multiplicity issues in clinical trials through adaptive hypothesis selection followed by group-sequential testing. Our proposed two-stage design begins with multiple hypotheses and adaptively selects a subset for further evaluation based on interim data. By using the group-sequential p-values and closed testing procedure, our approach effectively accounts for the group-sequential feature of the design and strongly controls the Type I error rate. The simulation study, motivated by a real oncology trial, demonstrates the efficacy of our approach in controlling the overall Type I error rate while preserving sufficient statistical power. By integrating adaptive design and multiplicity control principles, the methodology offers a practical and flexible framework for optimizing clinical trial designs, providing valuable insights for regulatory and clinical applications. However, several practical and methodological considerations warrant further discussion.

In our simulation study, we assumed that the first interim analysis (IA1) is conducted at month 15, with continuous patient enrollment from 0 to 3 years. In this setting, some patients enrolled close to the IA1 cutoff may not have sufficient follow-up time to assess PFS, and even fewer will have mature OS data. While a simple pause of enrollment before IA1 could help ensure that sufficient follow-up data inform interim decisions, it may also result in a loss of enrollment momentum or delay trial completion. These tradeoffs should be carefully evaluated during trial planning. To address this limitation, our methodology incorporates two strategies: (1) employing weighted test statistics in which the Stage-1 weights can be based on the planned number of events, allowing for reduced influence when event counts are low, and (2) leveraging the independent increment property of the log-rank test, whereby patients without events in Stage-1 still contribute to Stage-2 test statistics.

Regarding ORR-based decision-making at IA1, we implicitly assume that patients have been followed for a sufficient duration to assess ORR before the interim analysis. In practice, the overrun of enrollment from discontinued hypotheses may pose logistical challenges and potentially impact the interpretation of data, particularly in regulatory contexts. While such issues are beyond the scope of our current methodological framework, they are critical in real-world trial conduct. In addition, incorporating additional long-term follow-up data, such as PFS or OS, into the inference at the end of Stage-1 could enhance the efficiency and interpretability of the results. This is an important area for future methodological research.

According to the simulation results, the comparator design shows a slightly higher rejection probability than the proposed design in several scenarios. One potential explanation is that the Stage-1 selection criteria are based only on descriptive statistics of ORR, without any formal hypothesis testing, and we further assume that recruitment of T_2 in the BM+ population always continues into Stage-2. As shown in Figure 3, the weights for testing hypotheses other than H_1 and H_5 are stemmed from H_1 and H_5, and these hypotheses would only be tested if H_1 or H_5 are rejected. Consequently, if T_2 shows low efficacy compared to T_0 in the BM+ population, we may lose the power to test the remaining hypotheses. Further work is needed to improve both the selection criteria for adaptation and the weight assignment in the graphical procedure. Additionally, while our proposed design allows for pre-specified flexibility in treatment and population selection rules, the additional changes that may arise after IA1 were not initially anticipated. These can typically be addressed through protocol amendments. Such potential changes support the rationale for considering either a separate Phase 2 trial or an operationally seamless design, depending on the specific development and regulatory strategy.

Furthermore, while pre-specifying interim analyses based on planned sample size or number of events is important to maintain statistical validity, minor inflation in the Type I error rate may occur under data-driven or unplanned looks (Proschan, Lan, and Wittes 2006).

Finally, there are several other avenues for future research to enhance and generalize our proposed design and method. First, while the simulation results demonstrate favorable performance in terms of Type I error control and power, the method relies on pre-specified weights and error-spending functions. Future research could explore data-driven approaches to optimize these choices, such as incorporating adaptive weights in the inverse normal combination test (Luo and Quan 2024). Second, another promising direction involves extending the methodology to accommodate more complex adaptive designs, such as those with multiple interim looks or continuous adaptation strategies (Thall 2021). Third, although our work focuses on statistical methodology, evaluating potential cost savings of the proposed design is important for assessing its practical impact. While cost-effectiveness analyses are inherently trial-specific and operationally complex, future research could integrate economic considerations into the design framework.

Software

The relevant R code for the methodology and simulation study is available on https://github.com/kimihua1995/Adapt$_G$*roupSeq*.

Appendix Weighted Procedures

Suppose we have m hypotheses $H_1, \dots, H_m$ along with the corresponding non-negative weights $w_1, \dots, w_m$ such that $\sum_{j=1}^{m} w_j \leq 1$. We want to test the interception hypothesis $H = H_1 \cap \dots \cap H_m$ at level α based on the marginal p-values $p_1, \dots, p_m$.Testing of H can be conducted using the weighted Bonferroni procedure: reject H if $\min(p_1/w_1, \dots, p_m/w_m) \leq \alpha$.

Alternatively, testing of H can be done using the weighted parametric procedure: reject H if $\min(p_1/w_1, \dots, p_m/w_m) \leq \alpha C_H$, where $C_H \geq 1$ is determined via the joint distribution of $(p_1, \dots, p_m)$ under H.

Hochberg and Liberman (1994) proposed a weighted Simes procedure, denoted as HL, to conduct testing for H. Suppose $\sum_{j=1}^{m} w_j = 1$ and $\max(w_1, \dots, w_m) \leq 1/(m\alpha)$. Define q-value $q_j = p_j/w_j$, $j = 1, \dots, m$ and $q_{[1]} \leq \dots \leq q_{[m]}$ be the ordered q-values. Reject H if there exists some j such that $q_{[j]} \leq j\alpha$.

To remove the restriction on the weights, Benjamini and Hochberg (1997) proposed another weighted Simes procedure, denoted as BH, to conduct testing for H. Let $p_{[1]} \leq \dots \leq p_{[m]}$ be the ordered p-values and $w_{[1]}, \dots, w_{[m]}$ be the associated weights. Reject H if there exists some j such that $p_{[j]} \leq \alpha \sum_{i=1}^{j} w_{[i]}$.

Monotonicity of the Weighted Procedures

Suppose we have a new set of p-values $(p_1^*, \dots, p_m^*)$ which is componentwise not greater than $(p_1, \dots, p_m)$, denoted as $(p_1, \dots, p_m) \overset{c}{\geq} (p_1^*, \dots, p_m^*)$.

A procedure that rejects H based on $(p_1, \ldots, p_m)$ should reject H based on $(p_1^*, \ldots, p_m^*)$. We call this property the monotonicity of the procedure. Naturally, this is expected as smaller p-values should lead to a higher chance of rejection. This is indeed the case for the above weighted procedures as shown below.

The monotonicity of weighted Bonferroni and weighted parametric procedures is trivial. In the following, we will show the monotonicity of the two weighted Simes procedures. We only need to consider the situation when one and only one of the components of $(p_1, \ldots, p_m)$ is reduced.

To show the monotonicity of HL, let k be the index such that $q_{[k]} \le k\alpha$. We discuss the following scenarios:

(1) If one of $q_{[k+1]}, \ldots, q_{[m]}$ is reduced to q^* and

(1.1) if $q^* > q_{[k]}$, then $q^*_{[k]} = q_{[k]} \le k\alpha$;
(1.2) if $q^* = q_{[k]}$, then $q^*_{[k]} = q^*_{[k+1]} = q_{[k]} \le k\alpha$;
(1.3) if $q^* < q_{[k]}$, then $q^*_{[k+1]} = q_{[k]} \le k\alpha < (k+1)\alpha$.

(2) If one of $q_{[1]}, \ldots, q_{[k-1]}$ is reduced, then $q^*_{[k]} = q_{[k]} \le k\alpha$.
(3) If $q_{[k]}$ is reduced to q^* and

(3.1) if $q^* \ge q_{[k-1]}$, then $q^*_{[k]} \le q_{[k]} \le k\alpha$;
(3.2) if $q^* < q_{[k-1]}$, then $q^*_{[k]} = q_{[k-1]} \le q_{[k]} \le k\alpha$.

In other words, $q_{[k]} \le k\alpha$ implies $q^*_{[k]} \le k\alpha$. Therefore, HL has the monotonicity.

To show the monotonicity of BH, let $k = \min\{j : p_{[j]} \le \alpha \sum_{i=1}^{j} w_{[i]}\}$. We discuss the following scenarios:

(1) If one of $p_{[k+1]}, \ldots, p_{[m]}$ is reduced to p^* and

(1.1) if $p^* > p_{[k]}$, then $p^*_{[j]} = p_{[j]}$ and $w^*_{[j]} = w_{[j]}, j = 1, \ldots, k$, thus $p^*_{[k]} = p_{[k]} \le \alpha \sum_{i=1}^{k} w_{[i]} = \alpha \sum_{i=1}^{k} w^*_{[i]}$;
(1.2) if $p^* \le p_{[k]}$, then $p^*_{[k+1]} = p_{[k]}$ and the set $\{w_{[1]}, \ldots, w_{[k]}\}$ is a subset of the set $\{w^*_{[1]}, \ldots, w^*_{[k+1]}\}$, thus $p^*_{[k+1]} = p_{[k]} \le \alpha \sum_{i=1}^{k} w_{[i]} \le \alpha \sum_{i=1}^{k+1} w^*_{[i]}$;

(2) If one of $p_{[1]}, \ldots, p_{[k-1]}$ is reduced, then $p^*_{[k]} = p_{[k]} \le \alpha \sum_{i=1}^{k} w_{[i]} = \alpha \sum_{i=1}^{k} w^*_{[i]}$.
(3) If $p_{[k]}$ is reduced to p^* and

(3.1) if $p^* > p_{[k-1]}$, then $p^*_{[j]} = p_{[j]}, j = 1, \ldots, k-1$ and $w^*_{[j]} = w_{[j]}$, $j = 1, \ldots, k$, thus $p^*_{[k]} \le p_{[k]} \le \alpha \sum_{i=1}^{k} w_{[i]} = \alpha \sum_{i=1}^{k} w^*_{[i]}$;
(3.2) if $p^* = p_{[k-1]}$, then $p^*_{[k]} \le p_{[k]} \le \alpha \sum_{i=1}^{k} w_{[i]} = \alpha \sum_{i=1}^{k} w^*_{[i]}$;
(3.3) if $p^* < p_{[k-1]}$, $p^*_{[k]} = p_{[k-1]} \le p_{[k]} \le \alpha \sum_{i=1}^{k} w_{[i]} = \alpha \sum_{i=1}^{k} w^*_{[i]}$.

In summary, if there is a k such that $p_{[k]} \le \alpha \sum_{i=1}^{k} w_{[i]}$ then there must be a k^* such that $p^*_{[k^*]} \le \alpha \sum_{i=1}^{k^*} w^*_{[i]}$. Therefore, BH has the monotonicity.

Disclosure Statement

Dr. Luo is an employee of Sanofi and holds stock shares from the company; Dr. Wang serves on Data Monitoring Committee of several Sanofi trials; Dr. Hua declares no competing interests.

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